# PHOEBE CHEN

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## EDUCATION

**UNIVERSITY OF CALIFORNIA, LOS ANGELES** Expected Jun 2026

B.S. in **Statistics and Data Science** | Minor in **Data Science Engineering**

Relevant Coursework: Machine Learning, Neural Networks & Deep Learning, Causal Inference, Bayesian Statistics, Regression & Data Mining, Computational Statistics, Monte Carlo Methods, Linear Models, Database Management Systems

**MOORPARK COMMUNITY COLLEGE** Aug 2022 – May 2024

Honors Program

## PROJECTS

**Voter Turnout Analysis – GitHub** Jun 2026

- Built and compared XGBoost turnout classifiers on 33,315 Quincy voter records and 900 candidate variables; selected a compact 18-feature model achieving 0.803 ROC-AUC and 89.9% non-voter recall to prioritize 2027 municipal-election outreach.

**EMG-Based Keystroke Decoding with Neural Networks – GitHub** Mar 2026

- Designed and tuned a CNN-BiGRU model in PyTorch for EMG-based keystroke decoding, achieving approximately 16% character error rate through architectural improvements and hyperparameter optimization.

**Causal Analysis of Obesity and ICU Mortality** Mar 2026

- Estimated the effect of obesity on ICU mortality in MIMIC-IV using logistic regression and inverse probability weighting, finding approximately 15% lower mortality odds under the primary specification; assessed sensitivity and treatment-effect heterogeneity using alternative models and causal forests.

**LAPD Crime Data Analysis – GitHub** Mar 2026

- Analyzed 877K+ LAPD crime records in R using negative binomial, multinomial logistic, and chi-square models; estimated a 14% post-COVID increase in crime counts and produced reproducible visualizations with ggplot2, Quarto, and LaTeX.

**Heart Failure Mortality Prediction – GitHub** Dec 2025

- Built and compared logistic regression, random forest, and neural-network models for heart-failure mortality prediction, achieving 81.3% validation accuracy and identifying the most influential clinical risk factors.

## EXPERIENCE

**MOORPARK COLLEGE ENGINEERING CLUB** Sep 2022 – May 2024

Lead Programmer | Board Member – Secretary

- Led a 3-person programming team in qualifier competitions and the 2024 VEX U Robotics World Championship.
- Coordinated software design, hardware-software integration, and debugging in time-constrained settings.
- Developed autonomous navigation and control algorithms in C++ using PID controllers, improving autonomous task success rates by over 2x.

## SKILLS

**Programming:** Python, R, SQL, C++, MATLAB
**Libraries:** pandas, NumPy, scikit-learn, PyTorch, XGBoost, statsmodels, tidyverse, ggplot2
**Methods:** Regression, Classification, Causal Inference, Feature Engineering, Model Evaluation, Hyperparameter Tuning, EDA
**Tools:** Git, Jupyter, RStudio, Visual Studio, LaTeX, Quarto, Microsoft 365, Figma, Adobe Creative Cloud